

3.2.9	Alkenes	
	Alkenes: structure, bonding and reactivity	know that alkenes are unsaturated hydrocarbons
		know that bonding in alkenes involves a double covalent bond
		know that the arrangement $>C=C<$ is planar
		know that the alkenes can exhibit E-Z stereoisomerism
		be able to draw the structures of E and Z isomers
		understand that E-Z isomers exist due to restricted rotation about the C=C bond
		understand that the double bond in an alkene is a centre of high electron density
	Addition reactions of alkenes	understand the mechanism of electrophilic addition of alkenes with HBr, H_2SO_4 and Br_2
		know that bromine can be used to test for unsaturation
		be able to predict the products of addition to unsymmetrical alkenes by reference to the relative stabilities of primary, secondary and tertiary carbocation intermediates
		understand that alcohols are produced industrially by hydration of alkenes in the presence of an acid catalyst.
		know the typical conditions for the industrial production of ethanol from ethene
	Polymerisation of alkenes	know how addition polymers are formed from alkenes
		recognise that poly(alkenes) like alkanes are unreactive
		be able to recognise the repeating unit in a poly(alkene)
		know some typical uses of poly(ethene) and poly(propene) and know that poly(propene) is recycled